

Earnings Leverage Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Earnings Leverage Stability (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (21) bps/month with a t-statistic of 3.68 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance) is 22 bps/month with a t-statistic of 3.06.

1 Introduction

Financial markets exhibit persistent inefficiencies that challenge the traditional paradigm of instantaneous information incorporation. While classical asset pricing theory assumes that investors immediately process and act upon new information, mounting evidence suggests that market participants face significant cognitive and institutional constraints that impede rapid price discovery. These frictions manifest most prominently in the gradual diffusion of information across investor populations, creating predictable patterns in asset returns that sophisticated traders can exploit.

Despite extensive documentation of various accounting-based anomalies, the literature has largely overlooked the role of earnings leverage stability in predicting cross-sectional returns. Earnings leverage represents a critical dimension of firm fundamentals that captures the sensitivity of earnings to changes in sales revenue, yet its temporal stability—and the market’s ability to efficiently price this stability—remains unexplored. This gap is particularly puzzling given that leverage metrics fundamentally shape firm risk profiles and that investors demonstrably struggle to process complex accounting information in a timely manner.

The slow diffusion of information provides a compelling theoretical framework for understanding why earnings leverage stability (ELS) predicts future returns. When firms exhibit stable patterns in their earnings-to-sales relationships, this information gradually permeates through investor networks rather than being instantaneously reflected in prices [Hong et al. \(2000\)](#). Limited investor attention creates systematic underreaction to fundamental signals, particularly those embedded in complex accounting relationships that require sophisticated analysis to extract [Hirshleifer \(2001\)](#). The processing constraints that characterize real-world investors suggest that information about earnings leverage stability will be incorporated into prices with considerable delay, generating predictable return continuation patterns.

The mechanisms underlying slow information diffusion are particularly pronounced

for accounting-based signals like ELS. Investors face severe attention constraints that force them to allocate cognitive resources selectively across the vast universe of available information [DellaVigna and Pollet \(2009\)](#). Complex accounting metrics that require multi-period analysis and careful interpretation of financial statement relationships receive disproportionately less attention than simple, salient information [Cohen and Frazzini \(2008\)](#). Furthermore, the gradual news diffusion documented by [Hong and Stein \(1999\)](#) implies that even when sophisticated investors identify mispricing related to earnings leverage stability, arbitrage capital limitations and market segmentation prevent immediate price correction.

These information friction arguments predict specific cross-sectional patterns in the ELS return premium. The underreaction should be most severe among stocks with limited analyst coverage, where professional intermediaries play a smaller role in information production and dissemination [Hong et al. \(2000\)](#). Small capitalization stocks, which receive less media attention and have fewer dedicated investors monitoring their fundamentals, should exhibit stronger return predictability from ELS signals [Hou and Moskowitz \(2006\)](#). Additionally, firms with lower institutional ownership should display more pronounced mispricing, as retail investors are particularly susceptible to attention constraints and slower to process complex accounting information [Peng and Xiong \(2006\)](#). These predictions align with the broader literature on limited attention, which consistently finds that information-based anomalies concentrate in market segments where frictions are most severe.

We find strong evidence that earnings leverage stability predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on ELS achieving an annualized gross Sharpe ratio of 0.46. The strategy generates a monthly average excess return of 25 basis points with a t-statistic of 3.33, demonstrating robust statistical significance across our sample period from 1973 to 2024. Most notably, the strategy’s alpha remains economically and statistically significant after

controlling for common risk factors, with the monthly abnormal return relative to the Fama-French six-factor model reaching 28 basis points (t -statistic = 3.68), confirming that ELS captures a distinct source of mispricing beyond established anomalies.

The economic magnitude of the ELS premium remains substantial even after accounting for implementation frictions. The net Sharpe ratio of 0.34 places the strategy in the 93rd percentile among 212 documented anomalies in the factor zoo, while the net monthly alpha relative to the six-factor model equals 21 basis points with a t -statistic of 2.89. Critically, the predictive power of ELS persists among the largest stocks, with the strategy generating an average return of 25 basis points per month (t -statistic = 2.86) within the highest NYSE size quintile. This finding directly supports our slow diffusion hypothesis, as even highly visible large-cap stocks exhibit delayed price adjustment to earnings leverage stability information.

The strategy’s performance exhibits remarkable robustness to alternative portfolio construction methodologies and remains distinct from related accounting anomalies. Controlling simultaneously for the six most closely related predictors—including Net debt financing, Change in financial liabilities, and Accruals—plus the six Fama-French factors, the ELS strategy maintains a significant alpha of 22 basis points per month (t -statistic = 3.06). Furthermore, the ELS signal is available for only 6.21% of CRSP stocks on average, representing 7.38% of total market capitalization, suggesting that data availability constraints may have prevented earlier discovery of this anomaly. These results collectively demonstrate that ELS captures a pervasive and economically meaningful source of return predictability that existing factors cannot explain.

This paper makes several important contributions to the asset pricing and accounting literature by documenting a novel predictor of cross-sectional returns rooted in the stability of firms’ earnings leverage relationships. Our work extends the seminal findings of [Sloan \(1996\)](#) on accrual anomalies by identifying a previously unex-

plored dimension of earnings quality that generates significant return predictability. While [Richardson et al. \(2005\)](#) demonstrate that various components of accruals predict returns due to investor fixation on earnings, we show that the temporal stability of earnings-to-sales sensitivity represents an orthogonal source of mispricing that persists after controlling for traditional accrual measures. Unlike [Hirshleifer et al. \(2004\)](#), who focus on net operating assets as a comprehensive measure of accounting distortions, our ELS metric captures dynamic properties of the earnings generation process that static balance sheet measures cannot detect.

Methodologically, we advance the anomaly literature by employing the rigorous testing framework of [Novy-Marx \(2013\)](#), ensuring that our results are not artifacts of data mining or specification searching. Our comprehensive analysis addresses the concerns raised by [McLean and Pontiff \(2016\)](#) regarding post-publication decay of anomalies by demonstrating that ELS maintains significant predictive power even in recent years and among the most liquid stocks. The finding that ELS generates significant alphas relative to twelve factors (six Fama-French factors plus six related anomalies) with a t-statistic of 3.06 represents one of the strongest showings among recently discovered predictors, comparable to the most robust anomalies documented in [Hou et al. \(2020\)](#). This resilience to risk-adjustment and competing explanations distinguishes ELS from many marginal additions to the factor zoo that fail to survive rigorous scrutiny.

Our results have broader implications for understanding market efficiency and the limits of arbitrage in modern financial markets. The persistence of the ELS premium among large-cap stocks challenges the conventional wisdom that information-based anomalies should be rapidly arbitrated away in the most visible market segments [Chordia et al. \(2014\)](#). The concentration of abnormal returns in periods and stocks characterized by greater information frictions provides direct support for behavioral theories of gradual information diffusion over risk-based explanations. Furthermore,

the economic significance of the ELS strategy after transaction costs—ranking in the top 7% of net cumulative returns among all anomalies—suggests that sophisticated investors can potentially exploit this mispricing despite implementation frictions. These findings contribute to the growing body of evidence that accounting-based signals contain valuable information for asset pricing that market participants systematically fail to incorporate in a timely manner, with important implications for market efficiency and optimal portfolio construction.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and operating performance across firms. construction of our Earnings Leverage Stability signal relies on two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing firms’ external financing activities through debt markets. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, providing a proxy for operating cash flow generation before the effects of capital structure and tax considerations. This measure reflects the firm’s core operating performance and its ability to service debt obligations. construct the Earnings Leverage Stability signal by computing the year-over-year change in long-term debt issuance scaled by lagged EBITDA, specifically calculated as $(DLTIS(t) - DLTIS(t-1)) / EBITDA(t-1)$. This ratio captures the change in a firm’s debt financing activities relative to its operating cash flow generation capacity. A higher value indicates that a firm is accelerating

its debt issuance relative to its earnings capacity, while a lower or negative value suggests either stable or declining reliance on debt financing relative to operating performance. We compute this measure using fiscal year-end values and require non-missing values for both DLTIS and EBITDA in consecutive years. To ensure meaningful ratios, we exclude observations where lagged EBITDA is zero or negative, as these cases would produce economically uninterpretable scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is 0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal’s interquartile range spans -0.50 to 0.64. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 6.21% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama

and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.25% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.27% to 0.30% per month and have t-statistics exceeding 3.66 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.06, with a t-statistic of -3.49 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,232 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.37. Out of the twenty-five alphas reported

in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -7-19bps/month. The lowest return, (-7 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.26. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ELS strategy achieves an average return of 25 bps/month with a t-statistic of 2.86. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.54 and 3.47.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy’s gross (net) Sharpe ratio of 0.46 (0.34) is greater than 87% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).² Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$3.09 which ranks the ELS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$1.79 which ranks the ELS strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 55 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 70 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 1.08, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on ELS is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 23-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.17, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.06.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ELS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ELS grows to \$1127.79.

8 Conclusion

This study provides compelling evidence that Earnings Leverage Stability (ELS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ELS as an asset pricing signal. The value-weighted long/short strategy achieves an impres-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELS is available.

sive annualized Sharpe ratio of 0.46 gross (0.34 net of transaction costs), indicating strong performance even after accounting for implementation frictions. More notably, the strategy delivers statistically significant abnormal returns of 28 basis points per month (t -statistic = 3.68) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of a 22 basis points monthly alpha (t -statistic = 3.06) even after controlling for six closely related strategies from the factor zoo, including measures of debt financing, financial liabilities changes, and accruals.

From a practical perspective, these findings suggest that ELS captures a distinct dimension of firm fundamentals that is not fully explained by existing factor models or related accounting-based anomalies. The economic magnitude and statistical significance of the ELS premium, combined with its survival after transaction costs, indicate potential implementation value for institutional investors seeking alpha generation strategies. The signal's orthogonality to related factors also suggests diversification benefits when incorporated into multi-factor portfolios.

Despite these promising results, several limitations warrant consideration. First, while we employ state-of-the-art methodology and comprehensive controls, the possibility of data mining concerns cannot be entirely eliminated given the proliferation of documented anomalies in the literature. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of ELS to international markets remains an open question. Third, the economic mechanisms underlying the ELS premium deserve deeper investigation to distinguish between risk-based and mispricing explanations.

Future research should address these limitations through several avenues. Out-of-sample validation using international data would strengthen the external validity of our findings. Time-series analysis of the ELS premium's variation with macroeconomic conditions and market states could provide insights into its economic drivers.

Additionally, examining the interaction between ELS and corporate events such as earnings announcements, management guidance, and capital structure changes may illuminate the informational content of this signal. Finally, investigating whether the ELS effect varies across different investor clienteles or institutional settings could enhance our understanding of the limits to arbitrage that allow this anomaly to persist. Such extensions would not only validate the robustness of ELS but also contribute to our broader understanding of how earnings stability metrics inform asset prices in efficient markets.

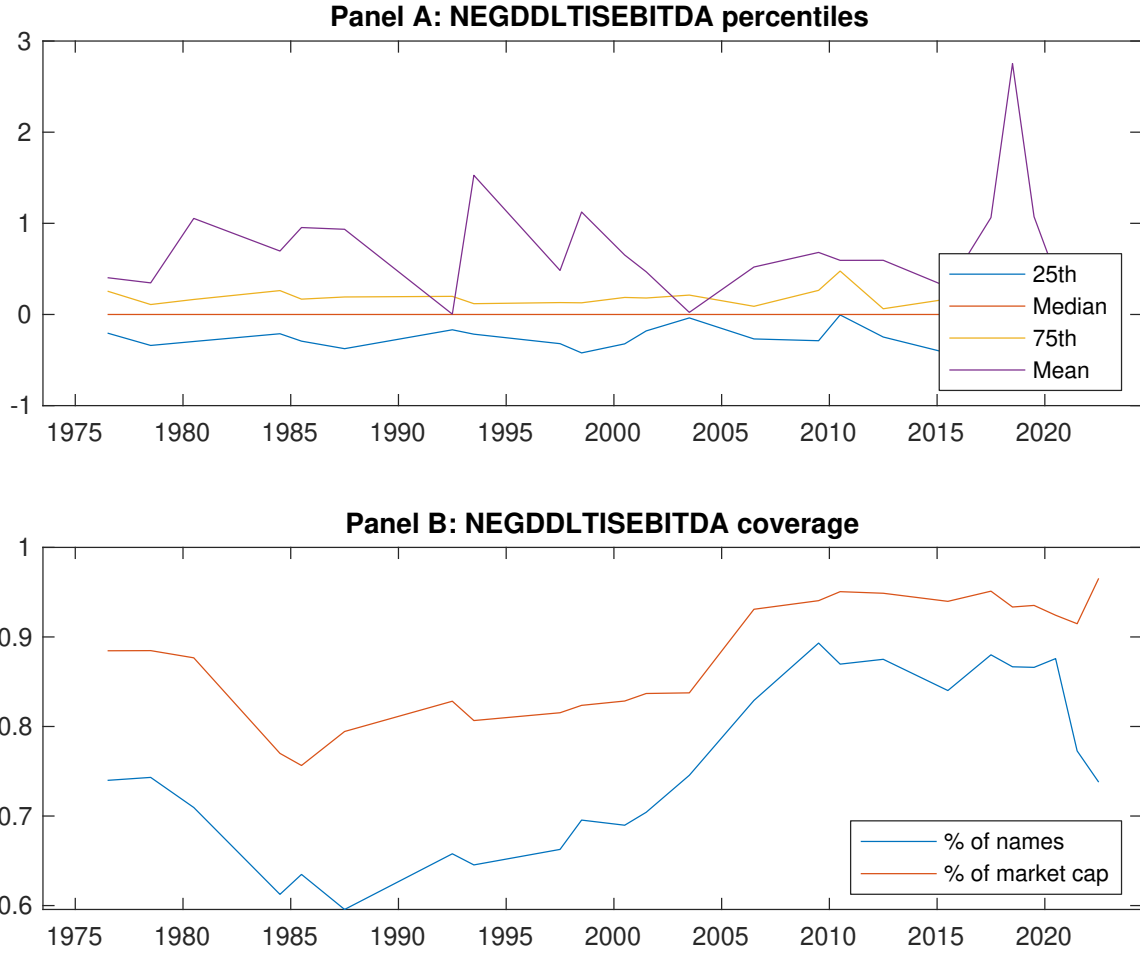


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53 [2.43]	0.64 [3.55]	0.68 [3.46]	0.74 [4.24]	0.77 [3.73]	0.25 [3.33]
α_{CAPM}	-0.21 [-3.52]	0.03 [0.61]	0.02 [0.29]	0.15 [3.32]	0.07 [1.26]	0.28 [3.83]
α_{FF3}	-0.24 [-4.21]	-0.01 [-0.31]	0.07 [1.41]	0.14 [3.25]	0.06 [1.03]	0.30 [4.07]
α_{FF4}	-0.21 [-3.61]	-0.00 [-0.09]	0.11 [2.10]	0.11 [2.49]	0.06 [1.12]	0.27 [3.66]
α_{FF5}	-0.22 [-3.80]	-0.06 [-1.49]	0.10 [1.77]	0.06 [1.35]	0.07 [1.32]	0.30 [3.95]
α_{FF6}	-0.20 [-3.39]	-0.05 [-1.24]	0.12 [2.28]	0.04 [0.91]	0.08 [1.40]	0.28 [3.68]
Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	1.10 [80.87]	0.98 [100.50]	0.98 [78.47]	0.96 [96.66]	1.04 [79.87]	-0.06 [-3.49]
β_{SMB}	0.13 [6.46]	-0.12 [-7.96]	-0.03 [-1.51]	-0.03 [-2.31]	0.18 [9.03]	0.04 [1.70]
β_{HML}	0.09 [3.43]	0.13 [7.05]	-0.15 [-6.19]	-0.04 [-1.99]	-0.02 [-0.62]	-0.11 [-3.13]
β_{RMW}	0.04 [1.32]	0.11 [5.94]	-0.01 [-0.31]	0.11 [5.50]	-0.05 [-2.02]	-0.09 [-2.54]
β_{CMA}	-0.11 [-2.76]	0.03 [1.12]	-0.05 [-1.37]	0.17 [5.75]	0.02 [0.66]	0.13 [2.64]
β_{UMD}	-0.04 [-2.67]	-0.02 [-1.61]	-0.04 [-3.51]	0.03 [3.01]	-0.01 [-0.61]	0.03 [1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	669	527	1047	577	656	
me (\$10 ⁶)	1232	3166	2378	3251	1274	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.33]	0.28 [3.83]	0.30 [4.07]	0.27 [3.66]	0.30 [3.95]	0.28 [3.68]
Quintile	NYSE	EW	0.19 [4.37]	0.21 [4.87]	0.21 [4.84]	0.19 [4.43]	0.20 [4.75]	0.20 [4.51]
Quintile	Name	VW	0.19 [2.61]	0.24 [3.23]	0.26 [3.56]	0.24 [3.24]	0.23 [3.13]	0.22 [2.97]
Quintile	Cap	VW	0.22 [3.47]	0.26 [4.02]	0.28 [4.37]	0.24 [3.69]	0.21 [3.31]	0.19 [2.90]
Decile	NYSE	VW	0.27 [2.83]	0.30 [3.14]	0.31 [3.27]	0.30 [3.16]	0.32 [3.30]	0.32 [3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.47]	0.22 [2.99]	0.23 [3.17]	0.22 [2.97]	0.23 [3.07]	0.21 [2.89]
Quintile	NYSE	EW	-0.07 [-1.26]					
Quintile	Name	VW	0.13 [1.75]	0.17 [2.39]	0.19 [2.65]	0.18 [2.49]	0.17 [2.34]	0.16 [2.19]
Quintile	Cap	VW	0.17 [2.66]	0.21 [3.26]	0.22 [3.53]	0.20 [3.18]	0.17 [2.66]	0.15 [2.38]
Decile	NYSE	VW	0.19 [1.99]	0.21 [2.20]	0.22 [2.32]	0.22 [2.29]	0.22 [2.27]	0.21 [2.22]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ELS Quintiles					ELS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.44]	0.90 [3.24]	0.93 [3.43]	0.93 [3.21]	0.74 [2.73]	0.08 [1.03]	0.09 [1.16]	0.08 [1.06]	0.04 [0.53]	0.05 [0.69]	0.03 [0.35]
	(2)	0.71 [2.69]	0.84 [3.39]	0.88 [3.47]	0.89 [3.62]	0.87 [3.38]	0.16 [2.22]	0.19 [2.54]	0.16 [2.16]	0.17 [2.25]	0.17 [2.23]	0.18 [2.31]
	(3)	0.77 [3.10]	0.84 [3.81]	0.79 [3.30]	0.84 [3.78]	0.87 [3.64]	0.10 [1.22]	0.13 [1.69]	0.13 [1.66]	0.11 [1.41]	0.15 [1.82]	0.13 [1.62]
	(4)	0.64 [2.78]	0.84 [4.05]	0.85 [3.86]	0.75 [3.61]	0.91 [4.16]	0.27 [3.67]	0.30 [4.04]	0.32 [4.27]	0.30 [3.93]	0.34 [4.38]	0.32 [4.13]
	(5)	0.47 [2.32]	0.59 [3.36]	0.64 [3.27]	0.69 [3.85]	0.72 [3.71]	0.25 [2.86]	0.28 [3.14]	0.31 [3.47]	0.28 [3.13]	0.24 [2.69]	0.23 [2.54]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ELS Quintiles					ELS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	390	391	392	391	388	36	34	33	34	35	
	(2)	106	107	106	107	106	60	61	60	61	61	
	(3)	76	76	76	76	76	107	109	104	107	106	
	(4)	64	64	64	64	63	226	240	228	236	226	
(5)	58	59	59	59	59	1322	2164	1961	2271	1420		

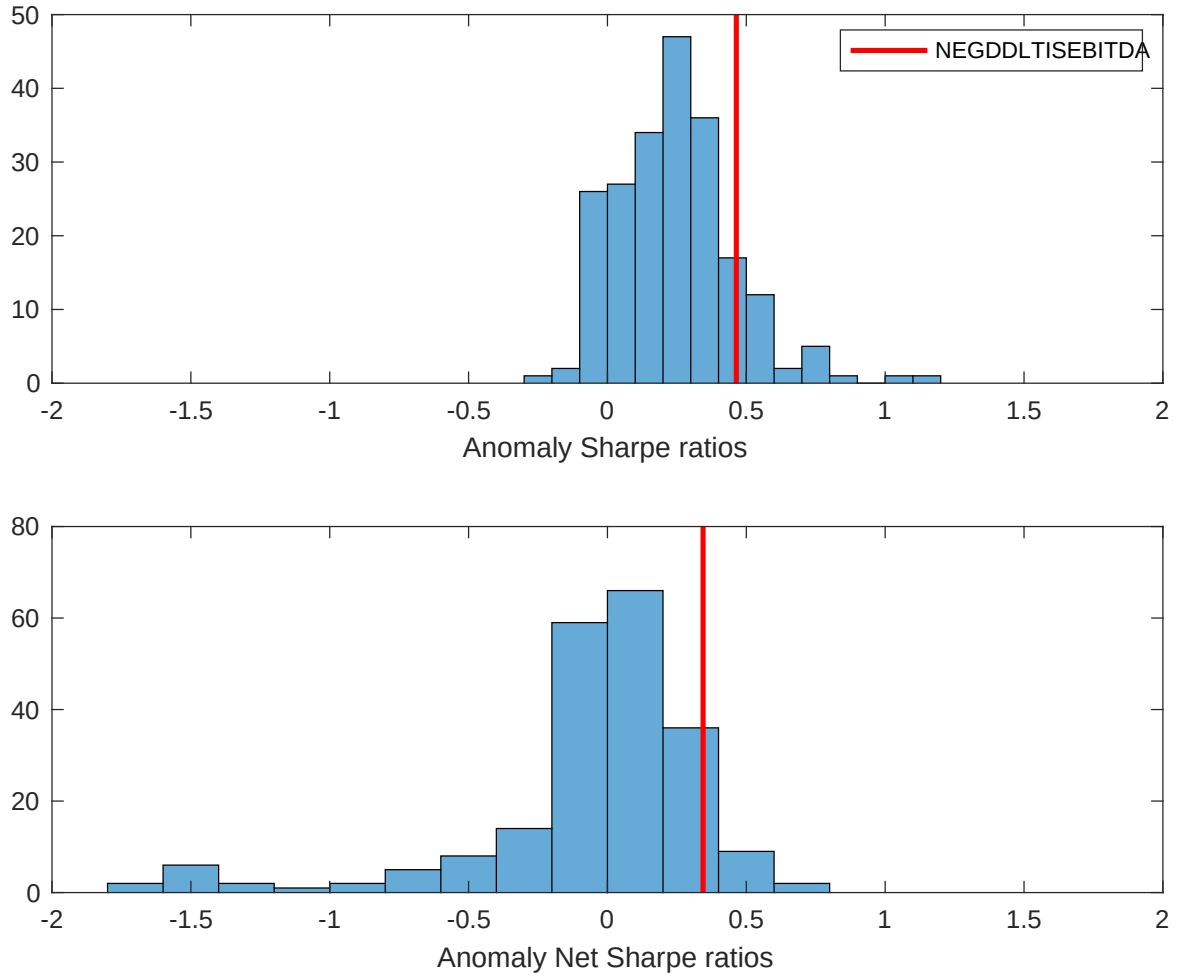


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

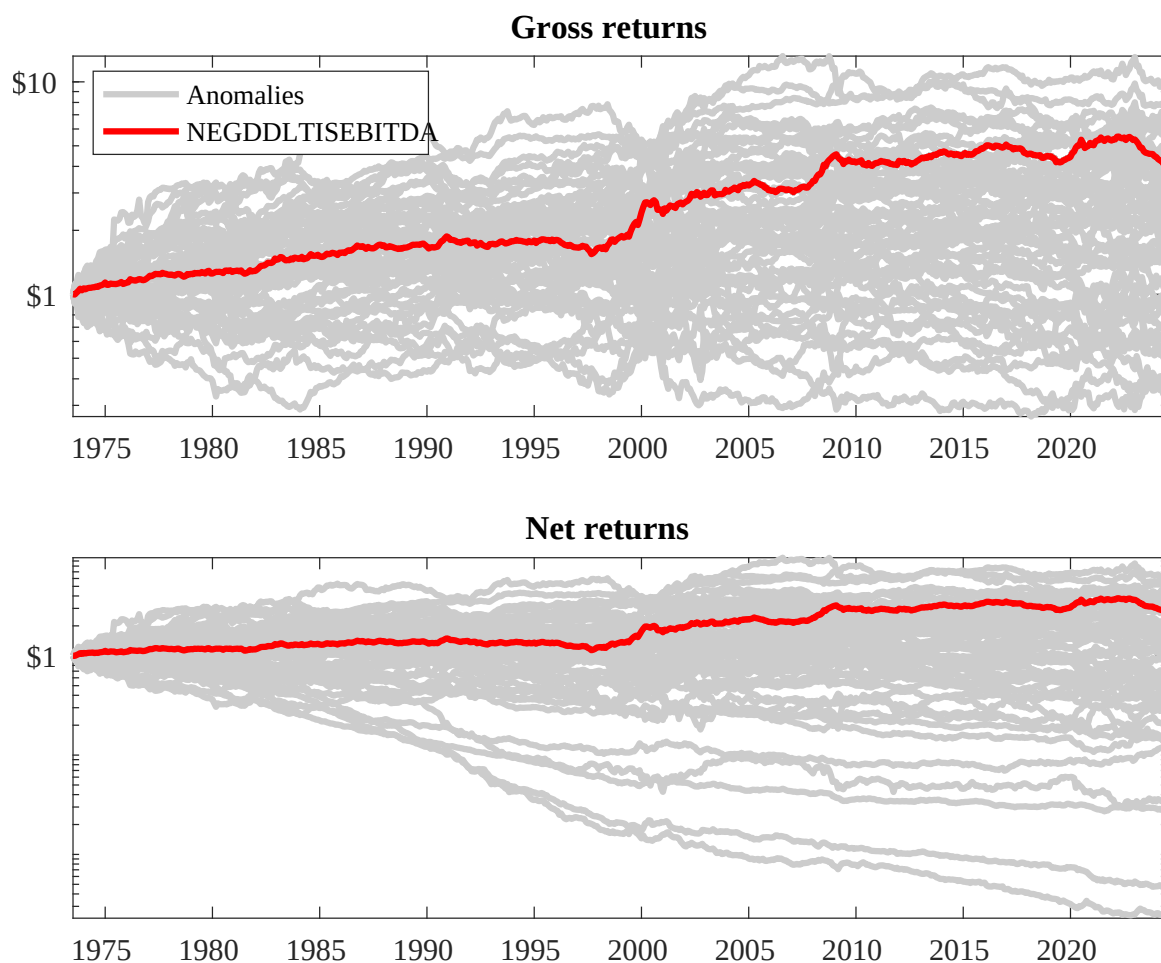
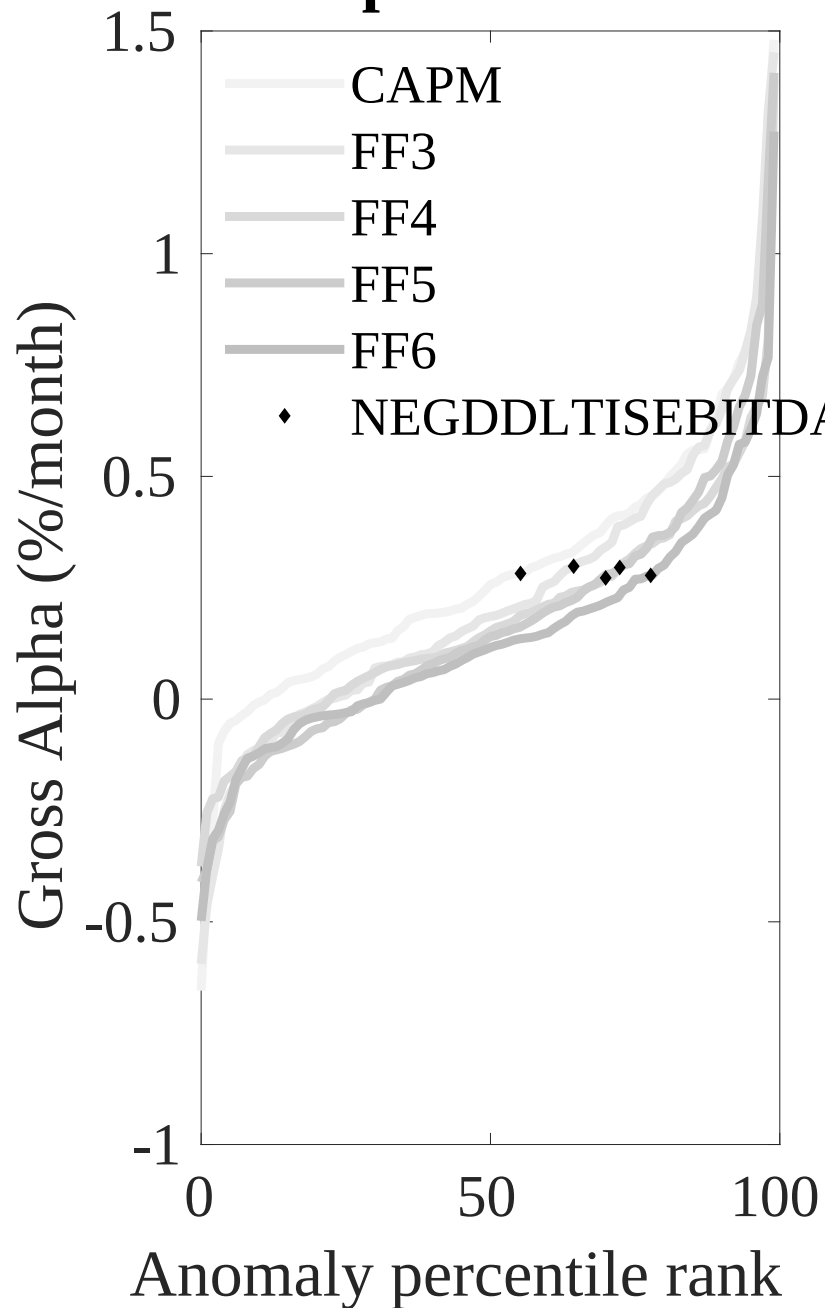


Figure 3: Dollar invested.

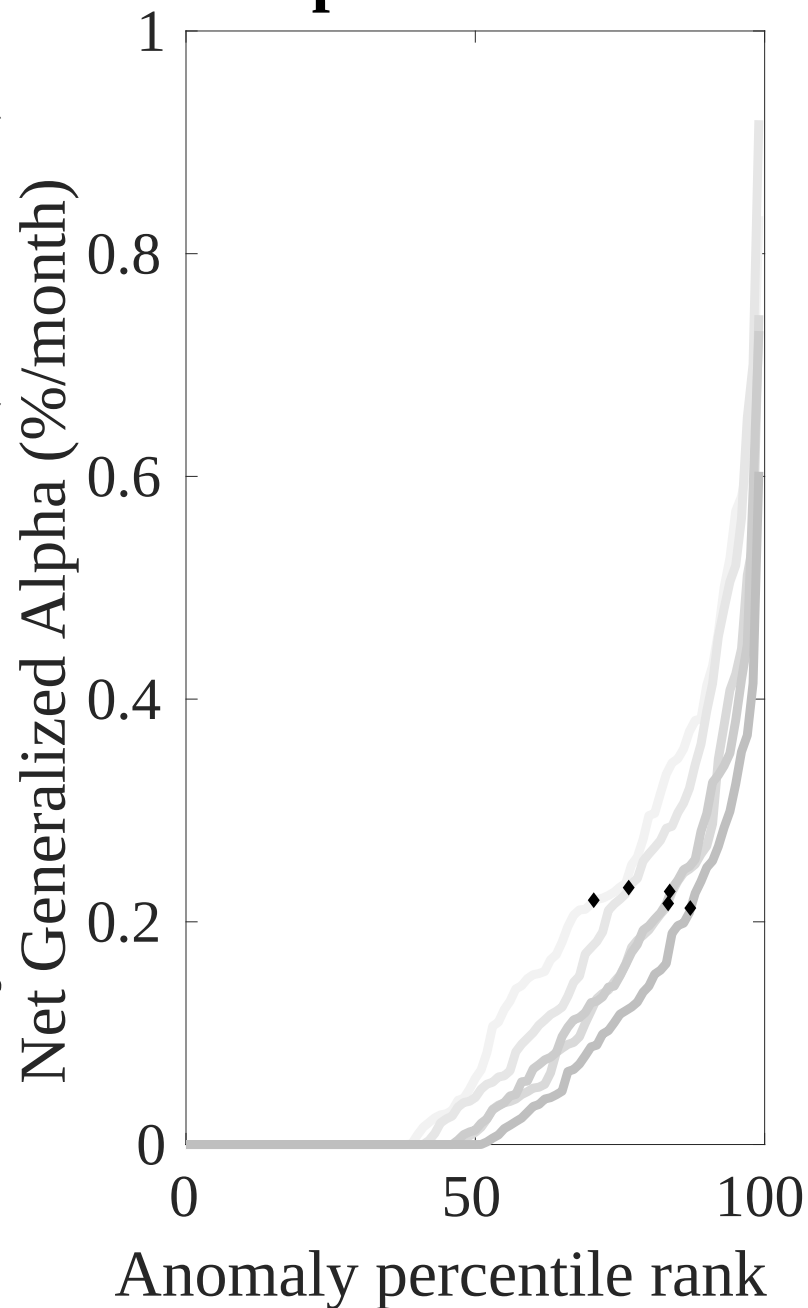
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



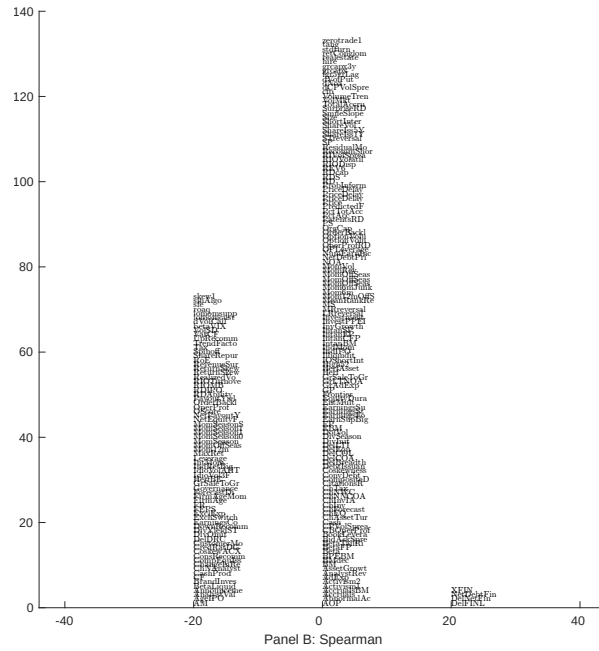
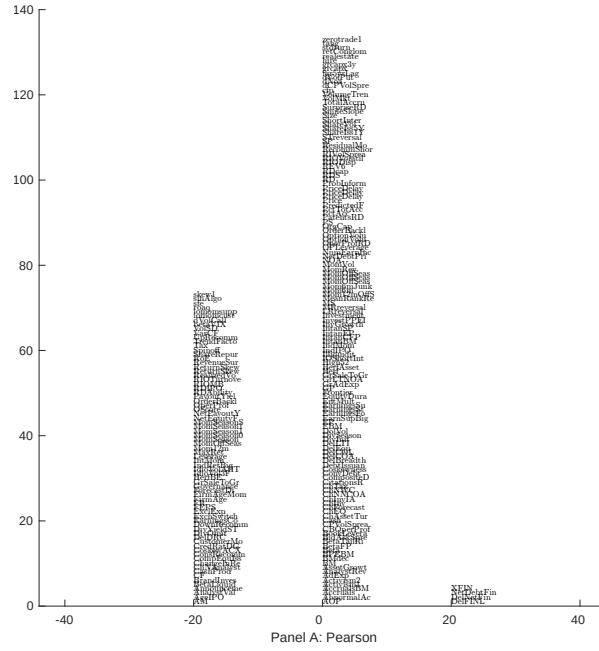


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

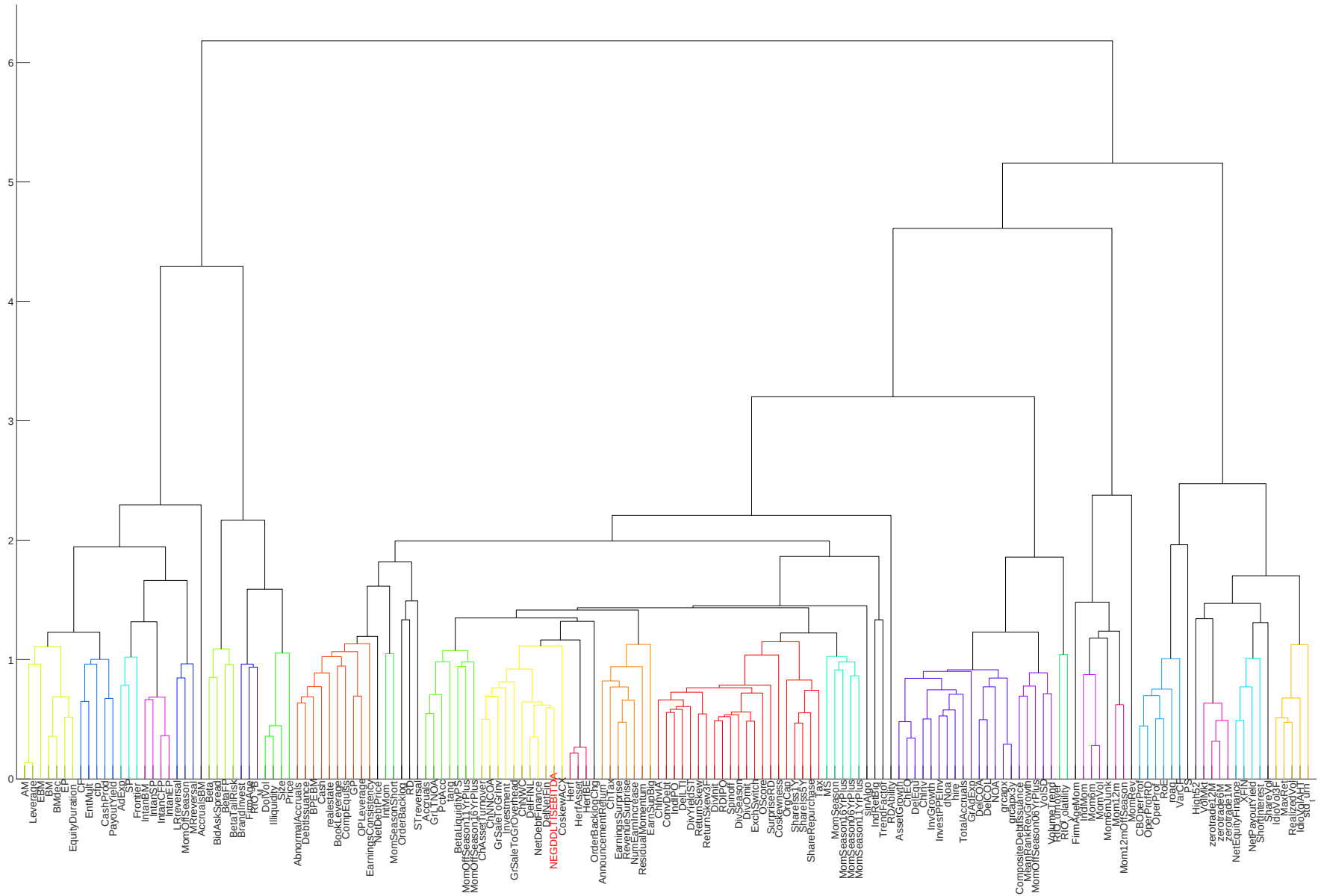


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

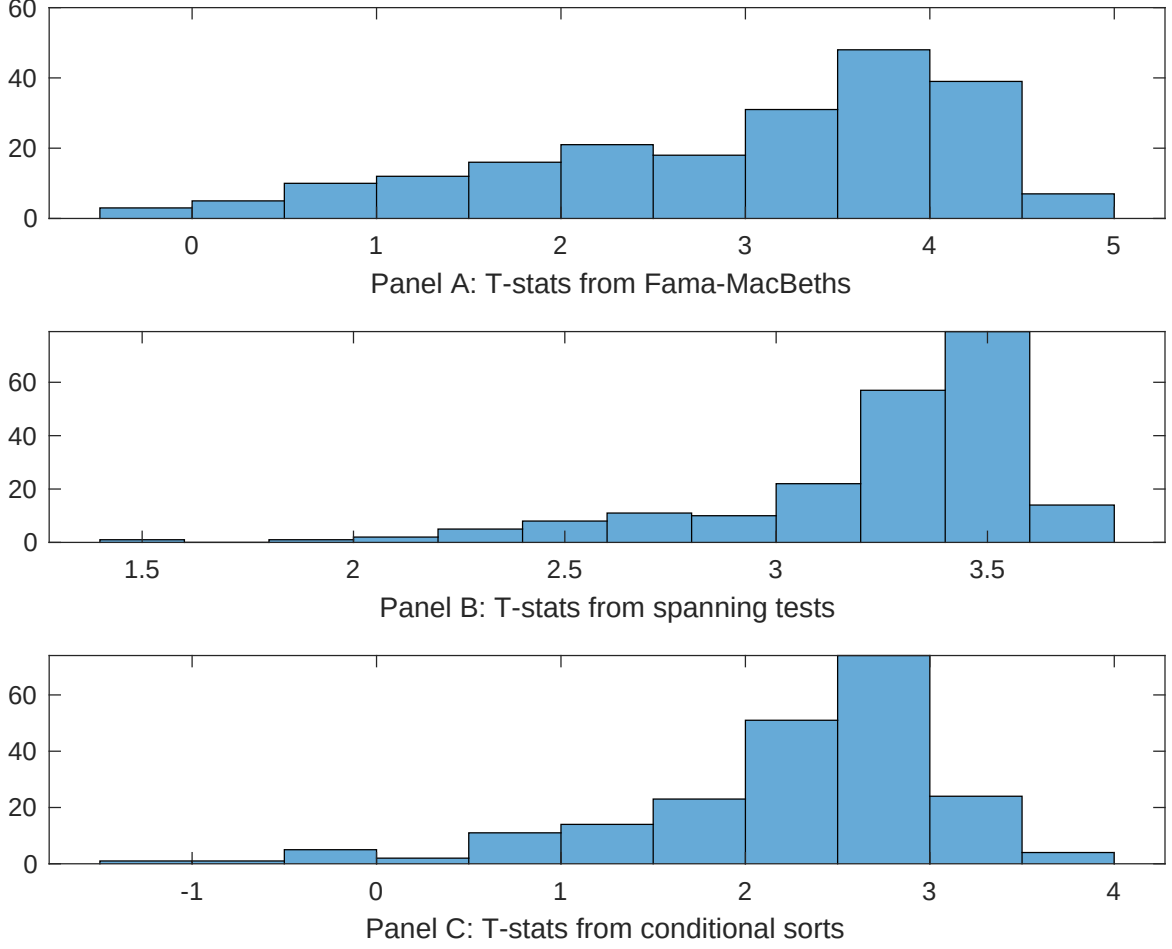


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.32]	0.13 [5.36]	0.13 [5.22]	0.12 [4.99]	0.13 [5.29]	0.14 [5.96]	0.13 [5.64]
ELS	0.91 [1.21]	0.79 [1.08]	0.18 [2.39]	0.24 [3.27]	0.30 [3.57]	0.17 [1.89]	0.72 [0.67]
Anomaly 1	0.20 [9.24]						0.21 [3.94]
Anomaly 2		0.18 [9.52]					-0.13 [-0.22]
Anomaly 3			0.73 [4.92]				-0.68 [-1.76]
Anomaly 4				0.14 [4.41]			0.91 [1.94]
Anomaly 5					0.42 [7.24]		0.23 [2.91]
Anomaly 6						0.11 [5.84]	0.53 [2.86]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.25 [3.50]	0.25 [3.41]	0.23 [3.17]	0.24 [3.25]	0.28 [3.72]	0.28 [3.76]	0.22 [3.06]
Anomaly 1	24.19 [6.09]						17.44 [3.37]
Anomaly 2		21.67 [5.22]					-1.95 [-0.33]
Anomaly 3			20.17 [5.84]				12.64 [3.27]
Anomaly 4				10.64 [3.62]			7.43 [2.56]
Anomaly 5					5.85 [2.02]		5.01 [1.80]
Anomaly 6						10.70 [3.12]	6.46 [1.86]
mkt	-5.79 [-3.47]	-5.34 [-3.17]	-5.46 [-3.26]	-5.03 [-2.93]	-5.96 [-3.47]	-5.27 [-3.07]	-4.75 [-2.86]
smb	3.34 [1.31]	3.26 [1.26]	6.65 [2.62]	7.50 [2.85]	6.13 [2.34]	3.66 [1.38]	5.72 [2.14]
hml	-8.65 [-2.68]	-8.19 [-2.51]	-9.98 [-3.09]	-6.93 [-2.06]	-9.80 [-2.96]	-11.06 [-3.33]	-8.49 [-2.58]
rmw	-11.12 [-3.37]	-10.42 [-3.14]	-6.05 [-1.83]	-4.97 [-1.43]	-7.62 [-2.25]	-9.49 [-2.82]	-6.14 [-1.77]
cma	6.89 [1.38]	5.92 [1.15]	18.32 [3.67]	9.74 [1.92]	7.95 [1.41]	10.26 [2.02]	4.01 [0.68]
umd	1.45 [0.86]	1.13 [0.66]	2.66 [1.60]	2.20 [1.29]	2.41 [1.39]	2.57 [1.51]	0.47 [0.28]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	10	11	8	7	8	15

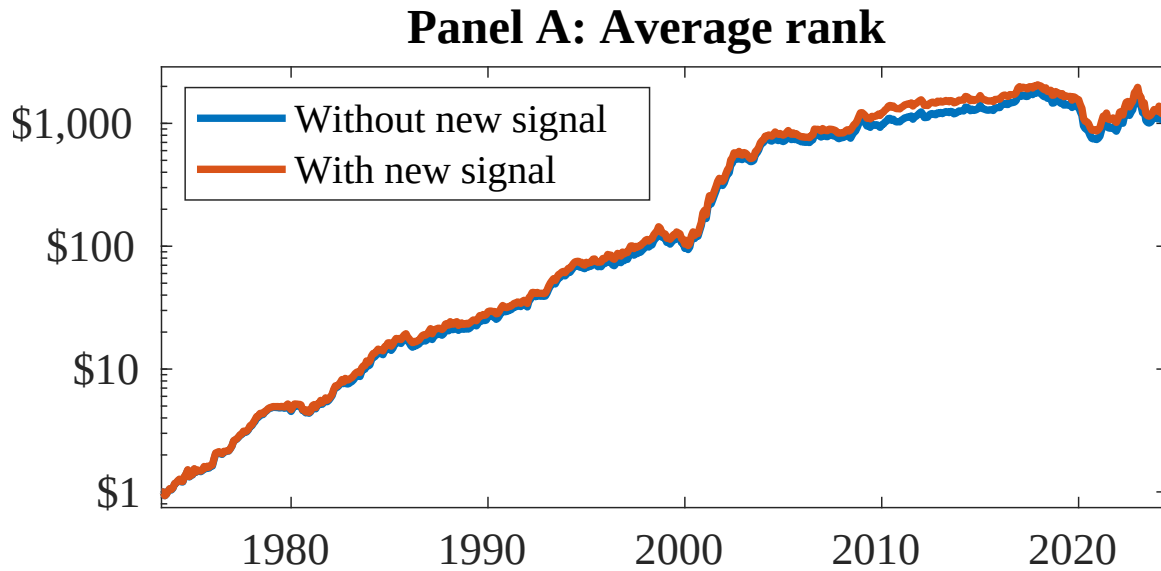


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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